

DETAILED DESCRIPTION OF THE INVENTION — SCOPE, VARIATIONS, AND DESIGN FORTIFICATION

The present disclosure is intended to encompass a broad range of structural, material, sensing, and operational variations associated with intraluminal physiologic monitoring devices. The embodiments described throughout this application are illustrative and should not be interpreted as limiting the scope of the invention.

In various embodiments, device geometries may include cylindrical bodies, tapered probes, curved bodies, segmented structures, flexible shafts, expandable assemblies, inflatable regions, articulated sections, modular components, or hybrid constructions combining rigid and compliant regions. Cross-sectional profiles may include circular, oval, elliptical, polygonal, asymmetric, adaptive, or dynamically deformable geometries designed to optimize comfort, retention, sensing fidelity, or manufacturability.

Sensor layouts may vary widely and may include circumferential rings, longitudinal strips, distributed arrays, matrix configurations, segmented sensing zones, clustered sensor regions, or dynamically reconfigurable sensing architectures. Sensors may be embedded within outer materials, positioned beneath protective layers, exposed through sensing windows, or integrated into flexible circuit structures.

Materials utilized in construction may include medical-grade silicone, elastomers, thermoplastic elastomers, polyurethanes, conductive polymers, composite materials, hydrogels, antimicrobial coatings, lubricious surface treatments, bio-compatible metals, ceramic components, or multilayer material systems. Material hardness, elasticity, surface texture, thermal conductivity, and compliance characteristics may be selected to achieve different functional objectives without departing from the disclosed invention.

Manufacturing techniques may include injection molding, overmolding, additive manufacturing, soft robotics fabrication, laminated flexible electronics assembly, automated robotic assembly, or hybrid fabrication approaches. Devices may be configured as reusable systems, disposable units, semi-disposable sensor cartridges, or upgradeable platform architectures.

Electronics placement may vary between distal, proximal, centralized, distributed, or externalized configurations. Wireless communication modules, processors, and power systems may be arranged to optimize signal integrity, thermal management, waterproofing, and ergonomic performance.

The disclosure further encompasses variations in insertion depth, orientation stabilization methods, retention features, calibration approaches, sensing activation sequences, and computational interpretation models. Future technological developments that achieve substantially similar physiologic monitoring outcomes are intended to fall within the scope of the invention.

Accordingly, the invention should be interpreted as covering all equivalents, substitutions, and modifications falling within the spirit and scope of the disclosed systems and methods, thereby establishing a comprehensive protective framework for intraluminal longitudinal pelvic physiology monitoring technologies.